CIRC HQ STUDENT HANDOUT | GRADE 5

Error-Proof Messages

Goal: Create a communication system that lets a receiver detect or correct a changed message.

35 minutes | 8 steps | Computer Science and Programming | Playbook B meeting 18, grade 5 path

Ask | Plan | Build | Test | Improve

You are done when

Show the altered message and explain exactly how the receiver detected or corrected it.

You will use

- Symbol cards
- Message strips
- Envelopes
- Check-rule cards
- Pencils

Steps

35 minutes total. Your teacher says when to move on.

1 Step 1: Get ready — 3 min

- Choose one symbol key.
- Place the channel envelope between teams.

2 Step 2: Message safety — 3 min

- Use no personal information.
- Use no secret information.
- Use respectful classroom words.

3 Step 3: Encode the message — 5 min

- Choose a short safe message.
- Convert it with the symbol key.
- Check every symbol once.

4 Step 4: Add an error check — 7 min

- Choose repeat, number-sum check, even-or-odd check, or pattern.
- Add the check to the message.
- Write the checking rule.

5 Step 5: Send through noise — 6 min

- Give the message to the change team.
- Let the channel change one part.
- Receive the changed message.

6 Step 6: Detect the error — 5 min

- Apply the checking rule.
- Mark the possible error.
- Decode the message.

7 Step 7: Reset the channel — 3 min

- Return every symbol card.
- Recycle used message strips.

8 Step 8: Exit check — 3 min

- Name the error check.
- State whether it found the change.

What to turn in

The changed message and how the receiver caught it.

No computer? Another way to do it

This lesson is already device free.

Finished early? Next challenge

Repair Mode: Improve the system so the receiver can correct one missing symbol, not only notice it.

Stay safe

Use no personal, insulting, or secret information in transmitted messages.

Stuck? Try this order

1. Read the step again, out loud if you can.
2. Ask a partner at your table.
3. Raise your hand. Keep working on the next part while you wait.